

Python Intermediate – Day 3 Checkpoint

Total Points: 16

Topics Covered

- Operator Overloading
 - Designing with Classes
 - Advanced Class Topics
 - Decorators & Class Gotchas
 - Software Engineering Best Practices with LLMs
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Student Information

Email: _____

Name: _____

Multiple Choice Questions

1. If you want to support the `+` operator for a custom class, which special method must you implement?

- A. `__plus__`
 - B. `__add__`
 - C. `__sum__`
 - D. `__concat__`
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2. Which special method enables indexing, such as `obj[0]`?

- A. `__index__`
- B. `__getitem__`
- C. `__access__`
- D. `__slice__`

3. To override the `<` operator, which special method should you implement?

- A. `__less__`
 - B. `__lt__`
 - C. `__cmp__`
 - D. `__compare__`
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4. What is the main advantage of composition over inheritance?

- A. Faster execution
 - B. Less memory usage
 - C. Better flexibility and loose coupling
 - D. Automatic polymorphism
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5. An "is-a" relationship represents:

- A. Composition
 - B. Delegation
 - C. Inheritance
 - D. Factory Pattern
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6. Why is an Abstract Base Class (ABC) used in Python? (1 point)

- A. To improve program speed
 - B. To store abstract data types
 - C. To enforce that subclasses implement required methods
 - D. To prevent inheritance
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7. What happens if you try to instantiate an abstract class that contains an unimplemented `@abstractmethod`? (1 point)

- A. The object is created successfully.
- B. Python automatically implements the method.
- C. A `TypeError` is raised.
- D. The program ignores the abstract method.

8. Which special method allows an object to be used in a **for loop? (1 point)**

- A. `__len__()`
 - B. `__getitem__()`
 - C. `__iter__()`
 - D. `__contains__()`
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9. What is the purpose of Python docstrings? (1 point)

- A. To increase execution speed
 - B. To automatically fix coding errors
 - C. To document classes, methods, and functions for users and developers
 - D. To replace comments completely
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10. What is the primary purpose of Human-in-the-Loop (HITL) in AI-assisted coding? (1 point)

- A. To make AI work without human involvement
 - B. To allow humans to review, correct, and improve AI-generated outputs
 - C. To replace software developers completely
 - D. To increase the number of AI models used
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11. In AI systems, what is a token? (1 point)

- A. A programming language keyword only
 - B. A unit of text, such as a word, subword, or punctuation, that the AI processes
 - C. A type of computer processor
 - D. A software license key
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12. Which feature allows attribute validation logic?

- A. Mixins
- B. Descriptors / Properties
- C. MRO (Method Resolution Order)
- D. Metaclasses

13. In UML, inheritance is represented by:

- A. A dashed arrow
- B. A solid line with a hollow triangle
- C. A double arrow
- D. A diamond shape

14. In object-oriented design, "cross-cutting concerns" typically include:

- A. Data structures
- B. Logging and authentication
- C. Inheritance trees
- D. Class variables

15. Why should you prefer getters and setters (or properties) in Python?

- A. They provide slower execution.
- B. Direct attribute access is unsafe.
- C. They support encapsulation and validation.
- D. They are required by Python.

16. Which of the following is an example of using AI in *Review Mode*? (1 point)

- A. Asking AI to create a new banking application from scratch
- B. Asking AI to explain Python syntax
- C. Asking AI to analyze existing code, identify bugs, and suggest improvements
- D. Asking AI to generate random passwords